

LAST-MARGIN GATING UNDER A TRUE TEMPORAL ENDPOINT DATA-QUALITY SHOCK

Anonymous authors

Paper under double-blind review

ABSTRACT

Checkpoint policies are often evaluated under a fixed training distribution. We instead impose a controlled temporal endpoint shock in synthetic sequence classification: labels remain clean until a predetermined subset is flipped during the final quarter of training. A last-margin alarm compares the final calibration loss with its running minimum, applies a threshold frozen on separate FIT seeds, and rolls deployment back from `last` to best validation. On disjoint hold-out seeds, the alarm detects all 64/64 shocked runs and flags none of the 64 healthy runs; rollback reduces mean outer loss by 0.3395 relative to `last`. However, every shocked trajectory reaches its minimum immediately before the shock, so the gate, best-val, and early-stop return identical outcomes. The experiment therefore supports contamination detection and rollback relative to endpoint deployment, not discovery of a checkpoint unavailable to best-val. Healthy $m \equiv 0$ follows from strictly decreasing trajectories, leaving realistic deployment false alarms unmeasured.

1 INTRODUCTION

A late event can invalidate an otherwise usable endpoint checkpoint. Static corruption does not isolate this failure: when corruption is active from the first epoch, a poor final checkpoint does not show that a previously clean candidate was later damaged. We construct the temporal case directly, ending a clean prefix immediately before a final-quarter label shock. In every shocked hold-out run, the checkpoint selected by best validation is recorded just before the shock begins (Figure 1).

We audit a simple deployment signal, the *last margin* $m = L_{11} - \min_{0 \leq t \leq 11} L_t$, where L_t is calibration loss after epoch t . The signal is read only at the end of the run. If m exceeds a threshold calibrated without hold-out access, the gate raises a contamination flag and deploys the best-validation checkpoint rather than `last`. Figure 1 shows actual healthy and shocked trajectories, not a schematic.

The experiment is pre-registered and deliberately narrow. Separate FIT seeds calibrate the threshold; disjoint hold-out seeds settle detection and checkpoint outcomes (Tables 1 and 7). The endpoint default is damaged, whereas the gate rolls each affected run back to the same pre-shock checkpoint selected by best-val and early-stop (Tables 2 and 3). This distinction—measurable value versus `last`, but no checkpoint advantage over existing selectors—sets the scope of the paper.

Our contributions are:

1. a timing-controlled endpoint-shock protocol with a frozen FIT/hold-out split;
2. a paired measurement of alarmed rollback versus `last`; and
3. an explicit boundary: the gate does not outperform best-val or early-stop here, and the healthy carrier does not measure realistic false alarms.

2 RELATED WORK

Weight averaging and checkpoint ensembling provide alternatives to endpoint deployment. Stochastic weight averaging broadens the solution region (Izmailov et al., 2018); exponential moving averages are widely used to stabilize targets and model states (Tarvainen & Valpola, 2017); and model

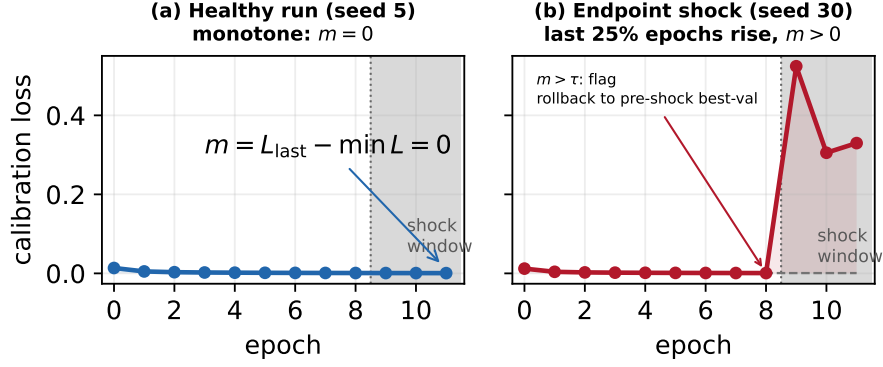


Figure 1: Actual E1 calibration trajectories. **Left:** healthy seed 5 decreases throughout, so the final point is the minimum and $m = 0$. **Right:** shocked seed 30 is clean through epoch 8; label flips activate only in the shaded epoch-9–11 window, lifting the trajectory above its pre-shock minimum. The frozen-threshold gate flags the run and returns that pre-shock best-val checkpoint. These representative runs were selected deterministically as closest to the across-run median trajectory; aggregate results use all 64 seeds per arm.

soups combine multiple fine-tuned checkpoints without additional inference cost (Wortsman et al., 2022). We include SWA, EMA, and soup as registered baselines, but our question is different: can a scalar read from the completed trajectory attest that the final endpoint was exposed to a late event?

Robustness work shows that nominal accuracy and robustness objectives can diverge (Tsipras et al., 2019). We use that citation only to motivate corruption as a stress test; it is not evidence about temporal label shocks, nor do we propose a label-noise correction method. Our contribution is the timing-controlled audit and its detection-versus-selection interpretation.

3 METHOD

3.1 A TRUE TEMPORAL ENDPOINT SHOCK

A run produces one calibration loss L_t and one model snapshot after each of $E = 12$ epochs. The shocked mode starts from the same clean training examples as the healthy mode. A predetermined subset of 150 of the 600 training examples is designated as affected. Crucially, designation does not change any label during epochs $0, \dots, 8$. The command-line control `--shock-time-frac=0.25` sets

$$t_{\text{shock}} = \lceil (1 - 0.25)E \rceil = 9. \quad (1)$$

During epochs 9, 10, 11 only, the affected examples use the deterministic target map $y \mapsto (y + 1) \bmod K$; all other labels remain clean. Epoch shuffling changes batch order but not this onset. The intervention is therefore a time-gated late-training event, rather than a claim that sample identity encodes training time.

The observed timeline validates the intended mechanism. All 64 shocked trajectories attain their minimum at epoch 8, immediately before $t_{\text{shock}} = 9$, and all subsequent shocked points remain above that minimum. Thus the clean candidate that later becomes best-val exists before the intervention, while `last` is recorded after three affected epochs.

3.2 LAST-MARGIN GATE

Define

$$m = L_{E-1} - \min_{0 \leq t < E} L_t \geq 0. \quad (2)$$

Table 1: Registered E1 configuration. The two fractions refer to different objects: 25% of examples are designated as affected, and their labels change only during the final 25% of epochs.

Component	Setting
Task	5 classes; vocabulary 400; sequence length 12; 6 informative features per class
Splits	train 600; calibration 200; outer 200; disjoint within each run
Model	3-layer transformer; hidden size 64; 4 heads; feed-forward size 128
Optimization	12 epochs; batch 32; learning rate 3×10^{-3} ; weight decay 0.01
Temporal shock	150 affected examples; labels clean in epochs 0–8 and flipped in epochs 9–11
Calibration	FIT seeds 101–104 in healthy and shocked modes; $\tau = 0.1354$
Settlement	hold-out seeds 1–64 in healthy and shocked modes
Selectors	last, best-val, early-stop, SWA, EMA, soup, rollback, margin-gate

The gate is

$$\text{gate}(m, \tau) = \begin{cases} \text{best-validation checkpoint,} & m > \tau, \\ \text{last checkpoint,} & m \leq \tau. \end{cases} \quad (3)$$

Firing is both an alarm and a rollback decision. The gate cannot return a checkpoint outside the pair $\{\text{last, best-val}\}$; consequently, its scientific value must be stated relative to the endpoint deployment it replaces, not as a new optimizer.

3.3 FROZEN CALIBRATION AND COMPARISON RULES

Let H and S be the healthy and shocked last margins from four FIT seeds in each mode. Before examining the hold-out settlement, we freeze

$$\tau = \frac{1}{2} (\max H + \min S) = 0.1354038956. \quad (4)$$

No hold-out trajectory is used to adjust this value. The registered comparison set contains the end-point, validation, stopping, averaging, soup, and rollback policies summarized with the full configuration in Table 1.

3.4 SETTLEMENT ESTIMANDS

The primary alarm endpoints are recall on shocked runs and false-alarm rate (FAR) on healthy runs, with one-sided 95% Clopper–Pearson bounds. Checkpoint value is measured on the shocked arm by the paired increment

$$\Delta_{\text{inc}} = \mathbb{E} [L_{\text{outer}}(\text{gate}) - L_{\text{outer}}(\text{last})]. \quad (5)$$

The positive control reverses the endpoints, $L_{\text{outer}}(\text{last}) - L_{\text{outer}}(\text{best-val})$, to verify that the late event actually damages endpoint deployment. We use the frozen paired bootstrap intervals and one-sided sign-flip permutation results in the settlement artifact; exact selector identity is reported as an equality rather than converted into a significance claim.

4 EXPERIMENTAL SETUP

4.1 CARRIER AND RUN INVENTORY

The carrier is a self-contained 5-class sequence-classification task trained on CPU. FIT and hold-out seeds are disjoint, and training, calibration, and outer splits are disjoint within each run. Table 1 moves the complete configuration out of prose; Table 7 in the appendix records every settlement unit.

4.2 PRE-REGISTRATION AND FROZEN DECISION RULE

The E1 pre-registration was frozen while only the first hold-out seed had completed. It declares four pass conditions: the healthy FAR upper bound is at most 0.10; at least one shocked run is detected; the paired confidence interval for gate-minus-last is wholly below zero; and the paired confidence interval for last-minus-best-val is wholly above zero. The settlement is a direct execution

Table 2: Mean outer outcomes over 64 hold-out seeds per mode. All rows are recomputed from the E1 raw run records. Bold marks the lowest shocked loss and highest shocked accuracy; equality is retained rather than broken by arbitrary ranking.

Rule	Healthy		Shocked	
	Outer loss	Outer accuracy	Outer loss	Outer accuracy
last	0.00063	1.00000	0.34046	0.95547
best-val	0.00063	1.00000	0.00092	1.00000
early-stop	0.00063	1.00000	0.00092	1.00000
SWA	0.00088	1.00000	0.09780	0.99227
EMA	0.00238	1.00000	0.03900	0.99531
soup	0.00164	1.00000	0.02812	0.99680
rollback	0.00070	1.00000	0.00106	1.00000
margin-gate	0.00063	1.00000	0.00092	1.00000

Table 3: Frozen E1 paired outer-loss settlement on the 64 shocked hold-out runs. The two bootstrap intervals were computed separately, so their endpoints are reported exactly as stored. Permutation $p \approx 0$ means zero counter-direction draws in the recomputed one-sided sign-flip settlement.

Contrast	Mean	95% CI	One-sided p
gate — last	−0.3395	[−0.3643, −0.3189]	≈ 0
last — best-val	+0.3395	[0.3187, 0.3634]	≈ 0
gate — best-val	0	exact equality in 64/64	not applicable
gate — early-stop	0	exact equality in 64/64	not applicable

of those conditions. We report all four, including the positive control that could have invalidated the temporal-shock premise.

5 RESULTS

5.1 THE ENDPOINT SHOCK DAMAGES `last`, WHILE PRE-SHOCK SELECTORS SURVIVE

Table 2 reports every registered checkpoint baseline on both modes and both outer metrics. Under the endpoint shock, `last` has much higher loss than `best-val` and the `gate`; its accuracy also falls while the pre-shock selectors remain perfect. Averaging baselines reduce but do not eliminate the loss increase. The table retains every exact value and tied optimum.

The `gate`, `best-val`, and `early-stop` outcomes are exactly equal in all 64 shocked records, for both outer loss and accuracy. Their common checkpoint is the epoch-8 snapshot. The alarm therefore restores an existing pre-shock candidate rather than creating a better checkpoint.

5.2 PAIRED ENDPOINT EFFECT AND ROLLBACK INCREMENT

Table 3 gives the two frozen paired contrasts. The positive control establishes the damage: `last` is worse than `best-val` by +0.3395. Turning on the `gate` relative to accepting `last` changes the same quantity with the opposite sign, −0.3395. Both intervals exclude zero by a wide margin, and the one-sided permutation estimates are approximately zero. The per-run positive controls are shown in Figure 2.

5.3 ALARM OPERATING CHARACTERISTICS

At the FIT-frozen $\tau = 0.1354$, every shocked margin exceeds the threshold and every healthy margin equals zero. Table 4 reports both point estimates and the pre-registered one-sided uncertainty bounds; Figure 3 shows the complete empirical distributions.

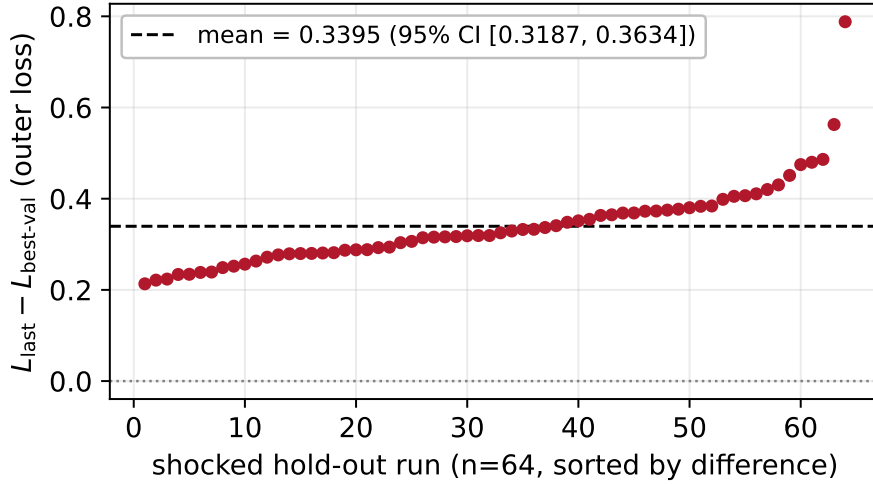


Figure 2: Per-run positive control $L_{\text{outer}}(\text{last}) - L_{\text{outer}}(\text{best-val})$ on 64 shocked hold-out runs, sorted. The mean is $+0.3395$ and the 95% CI is $[0.3187, 0.3634]$, read directly from the E1 settlement by the plotting script. Every observed difference is positive.

Table 4: Alarm operating characteristics at frozen $\tau = 0.1354$. Bounds are one-sided exact 95% Clopper–Pearson intervals in the conservative direction.

Endpoint	Count	Estimate	Exact bound
Recall on shocked runs	64/64	1.0	lower 0.9543
FAR on healthy runs	0/64	0.0	upper 0.0457

5.4 THE ZERO FALSE-ALARM RATE IS A DESIGN EXCLUSION, NOT DEPLOYMENT EVIDENCE

FAR = 0 is a real observation on this carrier, but the carrier excludes non-zero healthy margins. Every healthy trajectory is strictly decreasing at every adjacent transition, so its final point is necessarily the minimum and $m \equiv 0$. Table 5 distinguishes this mechanism from whole-cell readout collapse using the complete transition count and independent outer-loss and accuracy readouts.

The exact FAR bound therefore applies only to the registered monotone healthy population. A real fine-tuning carrier with normal stochastic non-monotonicity is required to measure the deployment false-alarm distribution; E1 does not supply that carrier.

5.5 INTERPRETATION AND BOUNDARY OF THE RESULT

5.5.1 THE FLAG HAS MEASURABLE ENDPOINT VALUE, BUT DOES NOT DISCOVER A NEW CHECKPOINT

Relative to the experimental deployment default, the flag prevents deployment of the damaged final snapshot, giving $\Delta_{\text{inc}} = -0.3395$. Best-val and early-stop require no alarm to obtain the same epoch-8 checkpoint. The supported claim is therefore *detection plus rollback relative to last*, not gate dominance over checkpoint selection.

The decisive next construction is E2: activate a shock early enough, or strongly enough, to suppress or displace the clean argmin so that best-val itself fails. If gate and best-val remain coextensive there, the checkpoint-novelty claim is falsified; if they separate under a frozen rule, that experiment would test a contribution E1 cannot establish.

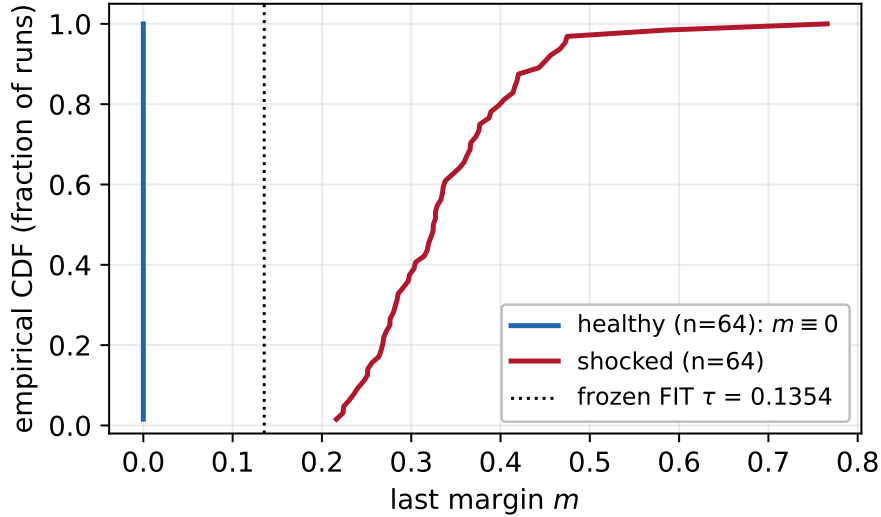


Figure 3: Empirical CDF of the last margin over all E1 hold-out runs. Healthy margins are exactly zero; shocked margins are dispersed but all exceed the FIT-frozen $\tau = 0.1354$. The separation supports the reported E1 operating point only; it is not a deployment calibration curve.

Table 5: Independent audit of the healthy $m \equiv 0$ cell. The observations distinguish a genuine monotone signature from an all-zero or non-responsive readout.

Diagnostic	Observed	Interpretation
Strict adjacent decreases	704/704	trajectory changes at every step monotone mechanism for $m = 0$
Argmin at final epoch	64/64	
Best-val outer loss	mean 0.000625	non-zero independent loss readout task solved, not a dead cell
Best-val outer accuracy	1.0 in 64/64	

6 DISCUSSION

Because the affected labels activate only after a clean prefix, the experiment establishes the ordering needed for an endpoint-shock audit: pre-shock minimum, shock onset, and damaged final deployment. The paired benefit and positive control in Table 3 quantify this same operational event from opposite directions.

The meaningful policy comparison is correspondingly narrow. The gate has measured value against `last`, but no checkpoint advantage over best-val or early-stop. SWA, EMA, soup, and rollback also absorb much of the late shock (Table 2), so the experiment does not establish that an explicit alarm is necessary when a pipeline already deploys a pre-shock selector.

Table 6 makes the evidential boundary explicit. In particular, the healthy result is internally valid but externally narrow: strict monotonicity is verified as a real signal, yet it removes the variation needed for a realistic FAR test.

7 LIMITATIONS

- Synthetic scale and intervention.** The carrier is a small CPU-trained classifier. The intervention flips a predetermined affected subset only in the final three epochs. This is a true late-training event, but it is not an external incident occurring after training has ended, and its effect size must not be transported to large-model fine-tuning.
- Healthy variation is excluded.** Table 5 verifies that healthy $m \equiv 0$ is a genuine monotone signature rather than readout collapse. Nevertheless, the absence of healthy non-

Table 6: Claim boundaries and decisive next evidence. A checked E1 observation is not promoted beyond the population or policy comparison that generated it.

Question	What E1 establishes	What remains open
Temporal integrity	Flips activate only in epochs 9–11	Other shock types and durations
Alarm detection	Recall 1.0 at frozen τ	Recall on a realistic carrier
Endpoint value	Gate beats <code>last</code> by 0.3395 loss	Gate advantage over best-val (E2)
False alarms	FAR 0.0 on verified monotone runs	FAR with natural non-monotonicity (E6)
Route-1 probe	Prior 36-run probe used a smaller, different carrier	No transport of its rates to E1

monotonicity makes $\text{FAR} = 0$ partly a design exclusion. E6 requires a realistic carrier before any deployment-safety claim.

3. **Best-val is not defeated.** Every shocked argmin occurs before shock onset, as shown in Figure 1, and gate, best-val, and early-stop therefore agree exactly. E1 measures the value of a flag relative to `last`; E2 must cover or suppress the argmin to test whether a gate can add checkpoint value beyond best-val.
4. **Threshold transport is untested.** The midpoint $\tau = 0.1354$ is blind to hold-out but calibrated on the same synthetic mechanism. No E1 threshold sweep is used to claim a safe band, and the frozen value is not a recommended deployment threshold.
5. **Secondary diagnostics use distinct carriers.** Route 3 is an in-sample overtraining diagnostic, and the prior Route-1 probe uses a downsized configuration. Neither supplies an E1 FAR estimate or a substitute for a realistic healthy carrier.

8 CONCLUSION

Under a controlled final-quarter label shock, the FIT-frozen last-margin alarm detects all shocked runs and rolls the damaged endpoint back to a pre-shock checkpoint. Its outer-loss benefit is measured relative to `last`; it offers no checkpoint advantage over best-val or early-stop, which return the same outcomes. Realistic false-alarm behavior and any advantage beyond existing selectors remain unmeasured.

REPRODUCIBILITY

The frozen pre-registration, FIT threshold, 128 raw hold-out records, settlement JSON, verdict, and deterministic figure script are included in the anonymous workspace. Table 7 enumerates the run budget. Quantitative figures are generated only from registered E1 records; the conceptual image-generation candidate described in the supplementary disclosure is not used as experimental evidence. AI assisted language editing; every claim and number was checked against the registered artifacts.

REFERENCES

- Pavel Izmailov, Dmitrii Podoprikin, Timur Garipov, Dmitry Vetrov, and Andrew Gordon Wilson. Averaging weights leads to wider optima and better generalization. In *Proceedings of the Thirty-Fourth Conference on Uncertainty in Artificial Intelligence (UAI)*, 2018. arXiv:1803.05407.
- Antti Tarvainen and Harri Valpola. Mean teachers are better role models: Weight-averaged consistency targets improve semi-supervised deep learning results. In *Advances in Neural Information Processing Systems 30 (NeurIPS)*, 2017. arXiv:1703.01780.
- Dimitris Tsipras, Shibani Santurkar, Logan Engstrom, Alexander Turner, and Aleksander Madry. Robustness may be at odds with accuracy. In *International Conference on Learning Representations (ICLR)*, 2019. arXiv:1805.12152.
- Mitchell Wortsman, Gabriel Ilharco, Samir Yitzhak Gadre, Rebecca Roelofs, Raphael Gontijo-Lopes, Ari S. Morcos, Hongseok Namkoong, Ali Farhadi, Yair Carmon, Simon Kornblith, and

Ludwig Schmidt. Model soups: averaging weights of multiple fine-tuned models improves accuracy without increasing inference time. In *Proceedings of the 39th International Conference on Machine Learning (ICML)*, 2022. arXiv:2203.05482.

A RUN INVENTORY AND PROVENANCE

Table 7: E1 run inventory. FIT runs only calibrate τ ; all point estimates and uncertainty reported for E1 use the 128 hold-out runs.

Phase	Seeds	Modes	Runs
FIT calibration	4	healthy, shocked	8
Hold-out settlement	64	healthy	64
Hold-out settlement	64	shocked	64
Total		136 (8 FIT + 128 hold-out)	

The temporal mechanism is implemented in `experiments/code/combined.py`; the frozen run and settlement procedure is in `experiments/code/run_e1_temporal_shock.sh`; the pre-registration is `experiments/prereg/PREREG_E1_TEMPORAL.md`; and the frozen numerical settlement is `results/three_arm_temporal/E1_SETTLEMENT_frac0.25.json`. The representative and aggregate figures are reproduced by `experiments/code/p5_e1_figs.py`.

B DISTINCT ROUTE-3 DIAGNOSTIC

This prior diagnostic tests a different failure mode and does not validate E1. On its 48 in-sample runs, positive last margin and low outer accuracy identify the same set, and last-margin and final-loss thresholds tie. The rates below are descriptive rather than deployment estimates.

Table 8: Route-3 in-sample overtraining diagnostic, reported separately from E1.

Statistic	Recall	FAR
last margin	1.0000	0.0000
final loss level	1.0000	0.0000
first-rise early-stop	0.8929	0.1000
non-monotonicity indicator	1.0000	0.1500

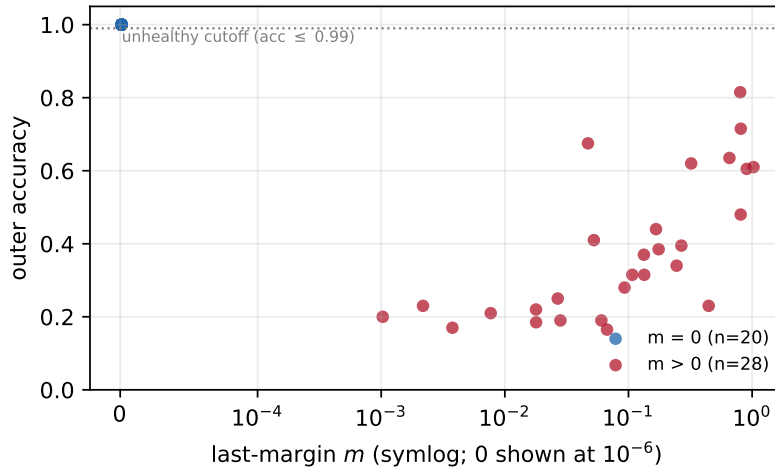


Figure 4: Route-3 overtraining scatter, retained as a distinct diagnostic. On these 48 runs, $m > 0$ and outer accuracy ≤ 0.99 identify the same 28 runs. The identity explains the perfect in-sample last-margin row in Table 8; it does not provide an E1 operating estimate.

C STATISTICAL DETAILS

Recall and FAR bounds are one-sided 95% Clopper–Pearson bounds. The two paired 95% confidence intervals in Table 3 use the bootstrap distributions frozen in the E1 settlement. The sign-flip permutation results were recomputed with 20,000 one-sided draws and correct elementwise signs; the stored empirical values are zero, which we report as $p \approx 0$ rather than as mathematical probability zero. Exact equality statements for gate, best-val, and early-stop come from recordwise equality of both stored outer loss and stored outer accuracy, not a null test.